

Jisun Ryu

jsryu01@g.skku.edu

Research Interests

Multi-omics integration for elucidating genetic architecture of disease phenotypes, Prioritization of therapeutic targets, Psychiatric genetics, Alzheimer's disease genetics, Complex trait genetics

Education

Sungkyunkwan University, Ph. D. course

School of Medicine

Mar. 2026 – Present

Seoul National University, M.P.H.

Graduate School of Public Health

Mar. 2024 – Feb. 2026

Kangwon National University, D.V.M.

College of Veterinary Medicine

Mar. 2018 – Feb. 2024

Research Experience

Genomics & Digital Health Laboratory, Sungkyunkwan University

Ph.D. course

Mar. 2026 – Present

Key Research:

- Exploring how phenotype definitions of psychiatric disorders influences its genetic architecture and biological interpretation through large-scale data analysis.
- Investigating genetic factors underlying treatment response and adverse events in Alzheimer's disease.

Genome & Health Big Data Laboratory, Seoul National University

Master's degree

Mar. 2024 – Feb. 2026

Key Research:

- Conducted proteomics-based Mendelian randomization (MR) studies to investigate causal relationships between lifestyle exposures, molecular traits, and complex diseases using UK Biobank (UKB) and UKB-PPP data.
- Investigated disease risk and causal relationships using survival analysis, Machine Learning, pathway enrichment analysis, and MR.
- Investigated the relationship between lifestyle exposures and depression across European and East Asian populations using UKB and KoGES data.

Publications

Myung, W.*, **Ryu, J*** & Won, HH. Balancing phenotyping depth and representativeness in psychiatric genomics. *Nature Genetetics* (2026). (co-first author)